Exercise 1.6

Q1:-

$$x_1 + x_2 = 2$$
$$5x_1 + 6x_2 = 9$$

$$\begin{bmatrix} 1 & 1 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$A \quad X = B$$
$$X = A^{-1}B$$

$$X = \left(\frac{1}{|A|}(Adj\,A)\right) \cdot B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{6-5}\begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

So, solution set = $\{3, -1\}$

Q2

$4x_1 -$
$2x_1 -$

$\begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$

Q2:—

$$4x_1 - 3x_2 = -3$$
$$2x_1 - 5x_2 = 9$$

$$\begin{bmatrix} 4 & -3 \\ 2 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$AX = B$$
$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\frac{1}{|A|}(\text{Adj } A)\right).B$$

$$= \frac{1}{-20+6}\begin{bmatrix} -5 & 3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$= \frac{1}{-14}\begin{bmatrix} -5 & 3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 5/14 & 3/14 \\ 1/7 & -2/7 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$+\frac{5}{14} - \frac{27}{14}$

$\frac{-42}{14} = -3$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \end{bmatrix}$$

$x_1 = -3$, $x_2 = -3$

Q3:—

$$x_1 + 3x_2 + x_3 = 4$$
$$2x_1 + 2x_2 + x_3 = -1$$
$$2x_1 + 3x_2 + x_3 = 3$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - R_3$, $R_3 - 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -1 \\ 0 & -3 & -1 & -2 & 0 & 1 \end{array}\right]$$

$(-1)\,R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & -3 & -1 & -2 & 0 & 1 \end{array}\right]$$

$R_3 + 3R_2$ =

$(-1)\,R_3$ =

$R_1 - 3R_2$

$R_3 - R_3$

$$R_3 + 3R_2 = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & -2 & -3 & 4 \end{array}\right]$$

$$(-1)R_3 = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$R_1 - 3R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & -3 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$R_1 - R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & -4 \end{bmatrix}$$

$$X = A^{-1} D$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 4 \\ -7 \end{bmatrix}$$

$x_1 = -1$

$x_2 = 4$

$x_3 = -7$

Q4:—

$5x_1 + 3x_2 + 2x_3 = 4$

$3x_1 + 3x_2 + 2x_3 = 2$

$0x_1 + x_2 + x_3 = 5$

$$\begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 5 & 3 & 2 & 1 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$-(R_1 - 2R_2) \left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - 3R_1$

$\left(-\frac{1}{6}\right)R_2$

$R_3 - R_2$

$(3)R_3$

$R_1 - 3R_2$

$R_2 - \frac{2}{3}R_3$

$$R_2 - 3R_1 \begin{bmatrix} 1 & 3 & 2 & | & -1 & 2 & 0 \\ 0 & -6 & -4 & | & 3 & -5 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$\left(-\frac{1}{6}\right)R_2 \begin{bmatrix} 1 & 3 & 2 & | & -1 & 2 & 0 \\ 0 & 1 & 2/3 & | & 1/2 & 5/6 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 - R_2 \begin{bmatrix} 1 & 3 & 2 & | & -1 & 2 & 0 \\ 0 & 1 & 2/3 & | & -1/2 & 5/6 & 0 \\ 0 & 0 & 1/3 & | & 1/2 & -5/6 & 1 \end{bmatrix}$$

$$(3)R_3 \begin{bmatrix} 1 & 3 & 2 & | & -1 & 2 & 0 \\ 0 & 1 & 2/3 & | & -1/2 & 5/6 & 0 \\ 0 & 0 & 1 & | & 3/2 & -5/2 & 3 \end{bmatrix}$$

$$R_1 - 3R_2 \begin{bmatrix} 1 & 0 & 0 & | & 1/2 & 1/2 & 0 \\ 0 & 1 & 2/3 & | & -1/2 & 5/6 & 0 \\ 0 & 0 & 1 & | & 3/2 & -5/2 & 3 \end{bmatrix}$$

$$R_2 - \frac{2}{3}R_3 \begin{bmatrix} 1 & 0 & 0 & | & 1/2 & 1/2 & 0 \\ 0 & 1 & 0 & | & -3/2 & 5/2 & -2 \\ 0 & 0 & 1 & | & 3/2 & -5/2 & 3 \end{bmatrix}$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1+0 \\ -6+5-10 \\ 6-5+15 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -11 \\ 16 \end{bmatrix}$$

Q5:—

$$x + y + z = 5$$
$$x + y - 4z = 10$$
$$-4x + y + z = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -4 \\ -4 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & 1 & 0 \\ -4 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} R_2 - R_1 \\ R_3 + 4R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -5 & -1 & 1 & 0 \\ 0 & 5 & 5 & 4 & 0 & 1 \end{array}\right]$$

$$R_2 \leftrightarrow R_3 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 5 & 5 & 4 & 0 & 1 \\ 0 & 0 & -5 & -1 & 1 & 0 \end{array}\right]$$

$$\left(\tfrac{1}{5}\right)R_2 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & -5 & -1 & 1 & 0 \end{array}\right]$$

$$\left(\tfrac{-1}{5}\right)R_3 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$R_1 - R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/5 & 0 & -1/5 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$R_2 - R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/5 & 0 & -1/5 \\ 0 & 1 & 0 & 3/5 & 1/5 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/5 & 0 & -1/5 \\ 3/5 & 1/5 & 1/5 \\ 1/5 & -1/5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/5 & 0 & -1/5 \\ 3/5 & 1/5 & 1/5 \\ 1/5 & -1/5 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 \\ 3+2+0 \\ 1-2+0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

Q6 :—

$$0w - x - 2y - 3z = 0$$
$$w + x + 4y + 4z = 7$$
$$w + 3x + 7y + 9z = 4$$
$$-w + 2x - 4y - 6z = 6$$

$$\begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 1 & 4 & 4 \\ 1 & 3 & 7 & 9 \\ -1 & -2 & -4 & -6 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \\ 4 \\ 6 \end{bmatrix}$$

$$A \quad X = B$$

$X =$

$A^{-1} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$

$R_1 \leftrightarrow R_2$

$R_3 - R_1$

$R_4 + R_1$

$\div R_2$

$R_3 - 2$

$R_4 +$

R_2

$X = A^{-1}B$

$$A^{-1} = \left[\begin{array}{cccc|cccc} 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & 6 & 0 & 0 & 0 & 1 \end{array}\right] \checkmark$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & -6 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_3 - R_1$, $R_4 + R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array}\right]$$

$-R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array}\right]$$

$R_3 - 2R_2$, $R_4 + R_2$, $R_1 - R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array}\right]$$

$(-1)R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array}\right]$$

$$\begin{array}{l} R_1 - 2R_3 \\ R_2 - 2R_3 \\ R_4 - 2R_3 \end{array} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 3 & -2 & 2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 & 2 & 1 \end{array}\right]$$

$$-R_4 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 3 & -2 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 & -2 & -1 \end{array}\right]$$

$$\begin{array}{l} R_2 + R_4 \\ R_2 - R_4 \\ R_3 - R_4 \end{array} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 2 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 6 & -3 & 4 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -3 & 1 & -2 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 2 & 0 & 0 & -1 \\ 6 & -3 & 4 & 1 \\ 1 & 0 & 1 & 1 \\ -3 & 1 & -2 & -1 \end{bmatrix}$$

$$X = A^{-1}B$$

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & -1 \\ 6 & -3 & 4 & 1 \\ 1 & 0 & 1 & 1 \\ -3 & 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 7 \\ 4 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -6 \\ 1 \\ 10 \\ -7 \end{bmatrix}$$

Q7:—

$$3x_1 + 5x_2 = b_1$$
$$x_1 + 2x_2 = b_2$$

$$\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{6-5} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{matrix} x_1 \\ x_2 \end{matrix} = \begin{bmatrix} 2b_1 - 5b_2 \\ -b_1 + 3b_2 \end{bmatrix}$$

Q8:—

$$x_1 + 2x_2 + 3x_3 = b_1$$
$$2x_1 + 5x_2 + 5x_3 = b_2$$
$$3x_1 + 5x_2 + 8x_3 = b_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 5 \\ 3 & 5 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = A^{-1} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 5 & 0 & 1 & 0 \\ 3 & 5 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & -1 & 1 & -3 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} R_1 - 2R_2 \\ R_3 + R_2 \\ \\ \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -2 & -5 & 1 & 1 \end{array}\right]$$

$$\begin{matrix} \\ \\ -\frac{1}{2}R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5/2 & -1/2 & -1/2 \end{array}\right]$$

$$R_2 + R_3 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -15/2 & 1/2 & 5/2 \\ 0 & 1 & 0 & 1/2 & 1/2 & -1/2 \\ 0 & 0 & 1 & 5/2 & -1/2 & 1/2 \end{array}\right]$$
$$R_1 - 5R_3$$

$$A^{-1} = \begin{bmatrix} -15/2 & 1/2 & 5/2 \\ 1/2 & 1/2 & -1/2 \\ 5/2 & -1/2 & 1/2 \end{bmatrix}$$

A

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -15/2 & 1/2 & 5/2 \\ 1/2 & 1/2 & -1/2 \\ 5/2 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

R

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{15}{2}b_1 + \frac{1}{2}b_2 + \frac{5}{2}b_3 \\ \frac{1}{2}b_1 + \frac{1}{2}b_2 - \frac{1}{2}b_3 \\ \frac{5}{2}b_1 - \frac{1}{2}b_2 - \frac{1}{2}b_3 \end{bmatrix}$$

Q9:—

(a-) if $b_1 = 1$, $b_2 = 4$

So, $x_1 - 5x_2 = 1$

$3x_1 + 2x_2 = 4$

In augmented matrix

$$\begin{bmatrix} 1 & -5 & : \\ 3 & 2 & : \end{bmatrix}$$

$R_2 - 3R_1$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\frac{1}{17} R_2$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$R_1 + 5R_2$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

So, $x_1 =$

(b-)

$R_2 - 3$

$\begin{bmatrix} 5/2 \\ -1/2 \\ 1/2 \end{bmatrix}$

$$\left[\begin{array}{cc:c} 1 & -5 & 1 \\ 3 & 2 & 4 \end{array}\right]$$

$R_2 - 3R_1$

$$\left[\begin{array}{cc:c} 1 & -5 & 1 \\ 0 & 17 & 1 \end{array}\right]$$

$\frac{1}{17} R_2$

$$\left[\begin{array}{cc:c} 1 & -5 & 1 \\ 0 & 1 & 1/17 \end{array}\right]$$

$R_1 + 5R_2$

$$\left[\begin{array}{cc:c} 1 & 0 & 22/17 \\ 0 & 1 & 1/17 \end{array}\right]$$

So, $x_1 = \frac{22}{17}$ $\quad$ $x_2 = \frac{1}{17}$

(b-) if $b_1 = -2$, $b_2 = 5$

$$x_1 - 5x_2 = -2$$
$$3x_1 + 2x_2 = 5$$

$$\left[\begin{array}{cc:c} 1 & -5 & -2 \\ 3 & 2 & 5 \end{array}\right]$$

$R_2 - 3R_1$

$$\left[\begin{array}{cc:c} 1 & -5 & -2 \\ 0 & 17 & 11 \end{array}\right]$$

$\frac{1}{17} R_2$

$$\left[\begin{array}{cc:c} 1 & -5 & -2 \\ 0 & 1 & 11/7 \end{array}\right]$$

$$R_2+5R_1 \begin{bmatrix} 1 & 0 & : & 21/17 \\ 0 & 1 & : & 11/17 \end{bmatrix}$$

So, $x_1 = \frac{21}{17}$, $x_2 = \frac{11}{17}$

Q10 :—

$$-x_1 + 4x_2 + x_3 = b_1$$
$$x_1 + 9x_2 - 2x_3 = b_2$$
$$6x_1 + 4x_2 - 8x_3 = b_3$$

$$\begin{bmatrix} -1 & 4 & 1 & : & b_1 \\ 1 & 9 & -2 & : & b_2 \\ 6 & 4 & -8 & : & b_3 \end{bmatrix}$$

(a-) if $b_1=0$, $b_2=1$, $b_3=0$

$$\begin{bmatrix} -1 & 4 & 1 & 0 \\ 1 & 9 & -2 & 1 \\ 6 & 4 & -8 & 0 \end{bmatrix}$$

$$(-1)R_1 \begin{bmatrix} 1 & -4 & -1 & 0 \\ 1 & 9 & -2 & 1 \\ 6 & 4 & -8 & 0 \end{bmatrix}$$

$$R_2-R_1 \begin{bmatrix} 1 & -4 \\ 0 & 13 \\ R_3-6R_1 & 0 & 28 \end{bmatrix}$$

$$\frac{1}{13}R_2 \begin{bmatrix} 1 & -4 \\ 0 & 1 \\ 0 & 2 \end{bmatrix}$$

$$R_1+4R_2 \begin{bmatrix} 1 \\ 0 \\ R_3-28R_2 & 0 \end{bmatrix}$$

$$\frac{13}{2}R_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$R_1+\frac{17}{13}R_3 \quad \begin{bmatrix}$$

$$R_2+\frac{1}{13}R_3$$

$x_1 = -18$

(b

$$\begin{matrix} R_2 - R_1 \\ R_3 - 6R_1 \end{matrix} \begin{bmatrix} 1 & -4 & -1 & 0 \\ 0 & 13 & -1 & 1 \\ 0 & 28 & -2 & 0 \end{bmatrix}$$

$$\frac{1}{13} R_2 \begin{bmatrix} 1 & -4 & -1 & 0 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 28 & -2 & 0 \end{bmatrix}$$

$$\begin{matrix} R_1 + 4R_2 \\ R_3 - 28R_2 \end{matrix} \begin{bmatrix} 1 & 0 & -17/13 & 4/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 2/13 & -20/13 \end{bmatrix}$$

$$\frac{13}{2} R_3 \begin{bmatrix} 1 & 0 & -17/13 & 4/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -14 \end{bmatrix}$$

$$\begin{matrix} R_1 + \frac{17}{13} R_3 \\ R_2 + \frac{1}{13} R_3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & -18 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -14 \end{bmatrix}$$

$x_1 = -18$, $x_2 = -1$, $x_3 = -14$

(b-) if $b_1 = -3$, $b_2 = 4$, $b_3 = -5$

$$\begin{bmatrix} -1 & 4 & 1 & -3 \\ 1 & 9 & -2 & 4 \\ 6 & 4 & -8 & -5 \end{bmatrix}$$

$$(-1)R_1 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 1 & 9 & -2 & 4 \\ 6 & 4 & -8 & -5 \end{bmatrix}$$

$$\begin{matrix} R_2 - R_1 \\ R_3 - 6R_1 \end{matrix} \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 13 & -1 & 1 \\ 0 & 28 & -2 & -23 \end{bmatrix}$$

$$\frac{1}{13}R_2 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 28 & -2 & -23 \end{bmatrix}$$

$$R_3 - 28R_2 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 2/13 & 327/13 \end{bmatrix}$$

$$\frac{13}{2}R_3 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & 327/2 \end{bmatrix}$$

$$R_1 + 4R_2 \begin{bmatrix} 1 & 0 & -17/13 & 43/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -327/2 \end{bmatrix}$$

$$R_1 + \frac{17}{13}R_3 \begin{bmatrix} 1 & 0 & 0 & 5645/26 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -327/2 \end{bmatrix}$$

$$R_2 + \frac{1}{13}R_3 \begin{bmatrix} 1 & 0 & 0 & 5645/26 \\ 0 & 1 & 0 & \\ 0 & 0 & 1 & 327/2 \end{bmatrix}$$

* (Complete yourself it).

$x_1 = \frac{-421}{2}$ $x_2 = \frac{-25}{2}$ $x_3 = \frac{-327}{2}$

Q 11:—

$4x_1 - 7x_2 = b_1$

$x_1 + 2x_2 = b_2$

$$\begin{bmatrix} 4 & -7 & b_1 \\ 1 & 2 & b_2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -7 : 0 : -4 : -1 : -5 \\ 1 & 2 : 1 : 6 : 3 : 1 \end{bmatrix}$$

(a-) (b-) (c-) (d-)

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 1 & 2 : 1 : 6 : 3 : 1 \\ 4 & -7 : 0 : -4 : -1 : -5 \end{bmatrix}$$

$R_2 - 4R_1$

$$\begin{bmatrix} 1 & 2 : 1 : 6 : 3 : 1 \\ 0 & -15 : -4 : 28 : -13 : -9 \end{bmatrix}$$

$\frac{-1}{15} R_2$

$$\begin{bmatrix} 1 & 2 : 1 : 6 : 3 : 1 \\ 0 & 1 : 4/15 : 28/15 : 13/15 : 3/5 \end{bmatrix}$$

$R_1 - 2R_2$

$$\begin{bmatrix} 1 & 0 & : 7/15 & : 34/15 & : 19/15 & : -1/5 \\ 0 & 1 & : 4/15 & : 28/15 & : 13/15 & : 3/5 \end{bmatrix}$$

(a-) $x_1 = 7/15$, $x_2 = 4/15$

(b-) $x_1 = 34/15$, $x_2 = 28/15$

(c-) $x_1 = 19/15$, $x_2 = 13/15$

(d-) $x_1 = -1/5$, $x_2 = 3/5$

Q12 :—

$$\begin{bmatrix} 1 & 3 & 5 & : b_1 \\ -1 & -2 & 0 & : b_2 \\ 2 & 5 & 4 & : b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 5 & : 1 & : 0 & : -1 \\ -1 & -2 & 0 & : 0 & : 1 & : -1 \\ 2 & 5 & 4 & : -1 & : 1 & : 0 \end{bmatrix}$$

(a-) (b-) (c-)

$R_2 + R_1$, $R_3 - 2R_1$

$$\begin{bmatrix} 1 & 3 & 5 & : 1 & : 0 & : -1 \\ 0 & 1 & 5 & : 1 & : 1 & : -2 \\ 0 & -1 & -6 & : -3 & : 1 & : 2 \end{bmatrix}$$

$R_1 - 3R_2$, $R_3 + R_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$(-1) R_3$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 - 5R_3$, $R_1 + 10R_3$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(a-)

(b-)

(c-)

R_2

$3/15 : -\frac{1}{5}$
$3/15 : \frac{3}{5}$

$$R_1 - 3R_2 \\ R_3 + R_2 \begin{bmatrix} 1 & 0 & -10 & : & -2 & : & -3 & : & 5 \\ 0 & 1 & 5 & : & 1 & : & 1 & : & -2 \\ 0 & 0 & -1 & : & -2 & : & 2 & : & 0 \end{bmatrix}$$

$$(-1)R_3 \begin{bmatrix} 1 & 0 & -10 & : & -2 & : & -3 & : & 5 \\ 0 & 1 & 5 & : & 1 & : & 1 & : & -2 \\ 0 & 0 & 1 & : & 2 & : & -2 & : & 0 \end{bmatrix}$$

$$R_2 - 5R_3 \\ R_1 + 10R_3 \begin{bmatrix} 1 & 0 & 0 & : & 18 & : & -23 & : & 5 \\ 0 & 1 & 0 & : & -9 & : & 11 & : & -2 \\ 0 & 0 & 1 & : & 2 & : & -2 & : & 0 \end{bmatrix}$$

(a-) $x_1 = 18$, $x_2 = -9$, $x_3 = 2$

(b-) $x_1 = -23$, $x_2 = 11$, $x_3 = -2$

(c-) $x_1 = 5$, $x_2 = -2$, $x_3 = 0$

Q13:—

$$\begin{bmatrix} 1 & 3 & b_1 \\ -2 & 1 & b_2 \end{bmatrix}$$

$$R_2 + 2R_1 \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 7 & b_2 + 2b_1 \end{bmatrix}$$

$$\frac{1}{7}R_2 \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & \frac{b_2 + 2b_1}{7} \end{bmatrix}$$

$R_1 - 3R_2$ $\begin{bmatrix} 1 & 0 & \frac{7b_1 - 3b_2 + 6b_1}{7} \\ 0 & 1 & \frac{b_2 + 2b_1}{7} \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & \frac{13b_1 - 3b_2}{7} \\ 0 & 1 & \frac{b_2 + 2b_1}{7} \end{bmatrix}$

System has unique solution

$x_1 = \frac{13b_1 - 3b_2}{7}$

$x_2 = \frac{b_2 + 2b_1}{7}$

Q14:—

$6x_1 - 4x_2 = b_1$

$3x_1 - 2x_2 = b_2$

$\begin{bmatrix} 6 & -4 & b_1 \\ 3 & -2 & b_2 \end{bmatrix}$

$\frac{1}{6}R_1$ $\begin{bmatrix} 1 & -2/3 & b_1/6 \\ 3 & -2 & b_2 \end{bmatrix}$

$R_2 - 3R_1$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

For system

$b_2 = \frac{b_1}{2}$

$b_2 =$

$\begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix}$

$R_2 - 4R_1$

$R_3 + 3R_2$

$R_3 + R_2$

syst

$R_2 - 3R_1$ $\begin{bmatrix} 1 & -2/3 & b_1/6 \\ 0 & 0 & b_2 - \frac{b_1}{2} \end{bmatrix}$

For system to be consistent

$$b_2 - \frac{b_1}{2} = 0$$

$$b_2 = \frac{b_1}{2} \qquad b_1 = 2b_2$$

Q15:—

$$\begin{bmatrix} 1 & -2 & 5 & b_1 \\ 4 & -5 & 8 & b_2 \\ -3 & 3 & -3 & b_3 \end{bmatrix}$$

$R_2 - 4R_1$, $R_3 + 3R_2$ $\begin{bmatrix} 1 & -2 & 5 & b_1 \\ 0 & 3 & -12 & b_2 - 4b_1 \\ 0 & -3 & 12 & b_3 + 3b_1 \end{bmatrix}$

$R_3 + R_2$ $\begin{bmatrix} 1 & -2 & 5 & b_1 \\ 0 & 3 & -12 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 + b_2 - b_1 \end{bmatrix}$

system is consistent because $R_3 = 0$.

$$b_3 + b_2 - b_1 = 0 \Rightarrow b_1 = b_3 + b_2$$

Q16 :—

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ -4 & 5 & 2 & b_2 \\ -4 & 7 & 4 & b_3 \end{bmatrix}$$

R_2+4R_1, R_3+4R_1

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & -3 & -2 & b_2+4b_1 \\ 0 & -1 & 0 & b_3+4b_2 \end{bmatrix}$$

$\frac{-1}{3}R_2$

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -\frac{(b_2+4b_1)}{3} \\ 0 & -1 & 0 & b_3+4b_1 \end{bmatrix}$$

R_3+R_2, R_2-

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -\frac{b_2-4b_1}{3} \\ 0 & 0 & 2/3 & \frac{8b_3-b_2+8b_1}{3} \end{bmatrix}$$

$\frac{3}{2}R_3$, R_2-R_3

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -4b_1-b_3 \\ 0 & 0 & 2/3 & \frac{3b_3-b_2+8b_1}{3} \end{bmatrix}$$

$\frac{3}{2}R_3$

$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{bmatrix}$$

R_1+2R_2

$$\begin{bmatrix} 1 & \\ 0 & 1 \\ 0 & \end{bmatrix}$$

R_3+R_3

$$\begin{bmatrix} 1 & \\ 0 & \\ 0 & \end{bmatrix}$$

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R_1+2R_2

$$\begin{bmatrix} 1 & 0 & -1 & -7b_1-2b_3 \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{bmatrix}$$

R_3+R_3

$$\begin{bmatrix} 1 & 0 & 0 & -6b_1-\frac{b_2}{2}-b_3 \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{bmatrix}$$

$$\therefore x_1 = \frac{-6b_1-b_2}{2} - b_3$$

$$x_2 = -4b_1 - b_3$$

$$x_3 = \frac{3b_3-b_2+8b_1}{2}$$

Q17:—

$$\begin{bmatrix} 1 & -1 & 3 & 2 & b_1 \\ -2 & 1 & 5 & 1 & b_2 \\ -3 & 2 & 2 & -1 & b_3 \\ 4 & -3 & 1 & 3 & b_4 \end{bmatrix}$$

$$\begin{array}{l|cccc|c} & 1 & -1 & 3 & 2 & b_1 \\ R_2+2R_1 & 0 & -1 & 11 & 5 & b_2+2b_1 \\ R_3+3R_2 & 0 & -1 & 11 & 5 & b_3+3b_1 \\ R_4-4R_1 & 0 & 1 & -11 & -5 & b_4-4b_1 \end{array}$$

$$\begin{array}{l|cccc|c} R_4+R_2 & 1 & -1 & 3 & 2 & b_1 \\ R_3-R_2 & 0 & -1 & 11 & 5 & b_2+2b_1 \\ & 0 & 0 & 0 & 0 & b_3+b_1-b_2 \\ & 0 & 0 & 0 & 0 & b_4+b_2-2b_1 \end{array}$$

So the system is consistant

If

$$b_1+b_2-2b_1=0$$

$$b_1=-b_2+b_3$$

$$2$$

Q18 :—

(a-)

$$AX=X$$

$$AX-IX=0$$

$$(A-I)X=0$$

$$So,\ \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{bmatrix},\quad X=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

So, $A-I=$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

So,

(A-I)

way.

R_2-2R_1

R_3-3R_2

$-R_2$

R_1-R_2

R_3+2

So, $A - I = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & -2 \\ 3 & 1 & 0 \end{bmatrix}$$

So, $(A-I)X = 0$ is formed in this way.

$$\begin{bmatrix} 1 & 1 & 2 & : & 0 \\ 2 & 1 & -2 & : & 0 \\ 3 & 1 & 0 & : & 0 \end{bmatrix}$$

$R_2 - 2R_1$, $R_3 - 3R_2$

$$\begin{bmatrix} 1 & 1 & 2 & : & 0 \\ 0 & -1 & -6 & : & 0 \\ 0 & -2 & -6 & : & 0 \end{bmatrix}$$

$-R_2$

$$\begin{bmatrix} 1 & 1 & 2 & : & 0 \\ 0 & 1 & 6 & : & 0 \\ 0 & -2 & -6 & : & 0 \end{bmatrix}$$

$R_1 - R_2$, $R_3 + 2R_1$

$$\begin{bmatrix} 1 & 0 & -4 & : & 0 \\ 0 & 1 & 6 & : & 0 \\ 0 & 0 & 6 & : & 0 \end{bmatrix}$$

$$\frac{1}{6}R_3 \begin{bmatrix} 1 & 0 & -4 & : & 0 \\ 0 & 1 & 6 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$\begin{matrix} R_1+4R_3 \\ R_2-6R_3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0$

$$X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(b-)

$AX = 4X$

$(A-4I)X = 0$

$$A-4I = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A-4I = \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}$$

$(A-4I)X =$

$$\begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}$$

$$-\frac{1}{2}R_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

R_2-2R_1

R_3-3R_1

$-R_2$

R_1+

R_3-

$R_2 - 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 5 & -2 & -2 & 1 & 0 \\ 0 & 2 & -1 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{5}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2/5 & -2/5 & 1/5 & 0 \\ 0 & 2 & -1 & 0 & 0 & 1 \end{array}\right]$$

$R_1 + R_2$, $R_3 - 2R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3/5 & 3/5 & 1/5 & 0 \\ 0 & 1 & -2/5 & -2/5 & 1/5 & 0 \\ 0 & 0 & -1/5 & 4/5 & -2/5 & 1 \end{array}\right]$$

$(-5)R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3/5 & 3/5 & 1/5 & 0 \\ 0 & 1 & -2/5 & -2/5 & 1/5 & 0 \\ 0 & 0 & 1 & -4 & 2 & -5 \end{array}\right]$$

$R_1 - \frac{3}{5}R_3$, $R_2 + \frac{2}{5}R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & 3 \\ 0 & 1 & 0 & -2 & +1 & -2 \\ 0 & 0 & 1 & -4 & 2 & -5 \end{array}\right]$$

$$X = A^{-1}B$$

$$= \left[\begin{array}{ccc} 3 & -1 & 3 \\ -2 & 1 & -2 \\ -4 & 2 & -5 \end{array}\right] \left[\begin{array}{ccccc} 2 & -1 & 5 & 7 & 8 \\ 4 & 0 & -3 & 0 & 1 \\ 3 & 5 & -7 & 2 & 1 \end{array}\right]$$

$$X_2 \begin{bmatrix} 11 & 12 & -3 & 27 & 26 \\ -6 & 8 & 1 & -18 & -17 \\ -15 & -21 & 9 & -38 & -35 \end{bmatrix}$$

Q 20 :—

$$\begin{bmatrix} -2 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 2 & -4 \end{bmatrix} X = \begin{bmatrix} 4 & 3 & 2 & 1 \\ 6 & 7 & 8 & 9 \\ 1 & 3 & 7 & 9 \end{bmatrix}$$

$$A \; X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} -2 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ -2 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

$R_3 + 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 2 & -7 & 1 & 0 & 2 \end{array}\right]$$

$-R_2$

$$\left[\begin{array}{ccc} 1 & 1 & -4 \\ 0 & 1 & 1 \\ 0 & 2 & -7 \end{array}\right.$$

$R_1 - R_2$
$R_3 - 2R_2$

$$\left[\begin{array}{ccc} 1 & 0 & \\ 0 & 1 & \\ 0 & 0 & \end{array}\right.$$

$-\frac{1}{9} R_3$

$$\left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array}\right.$$

$R_1 + 5R_3$

$$\left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array}\right.$$

$R_2 - R_3$

$$\left[\begin{array}{c} 1 \\ 0 \\ \end{array}\right.$$

$A^{-1} \cdot B_2$

$$-R_2 \begin{bmatrix} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 2 & -7 & 1 & 0 & 2 \end{bmatrix}$$

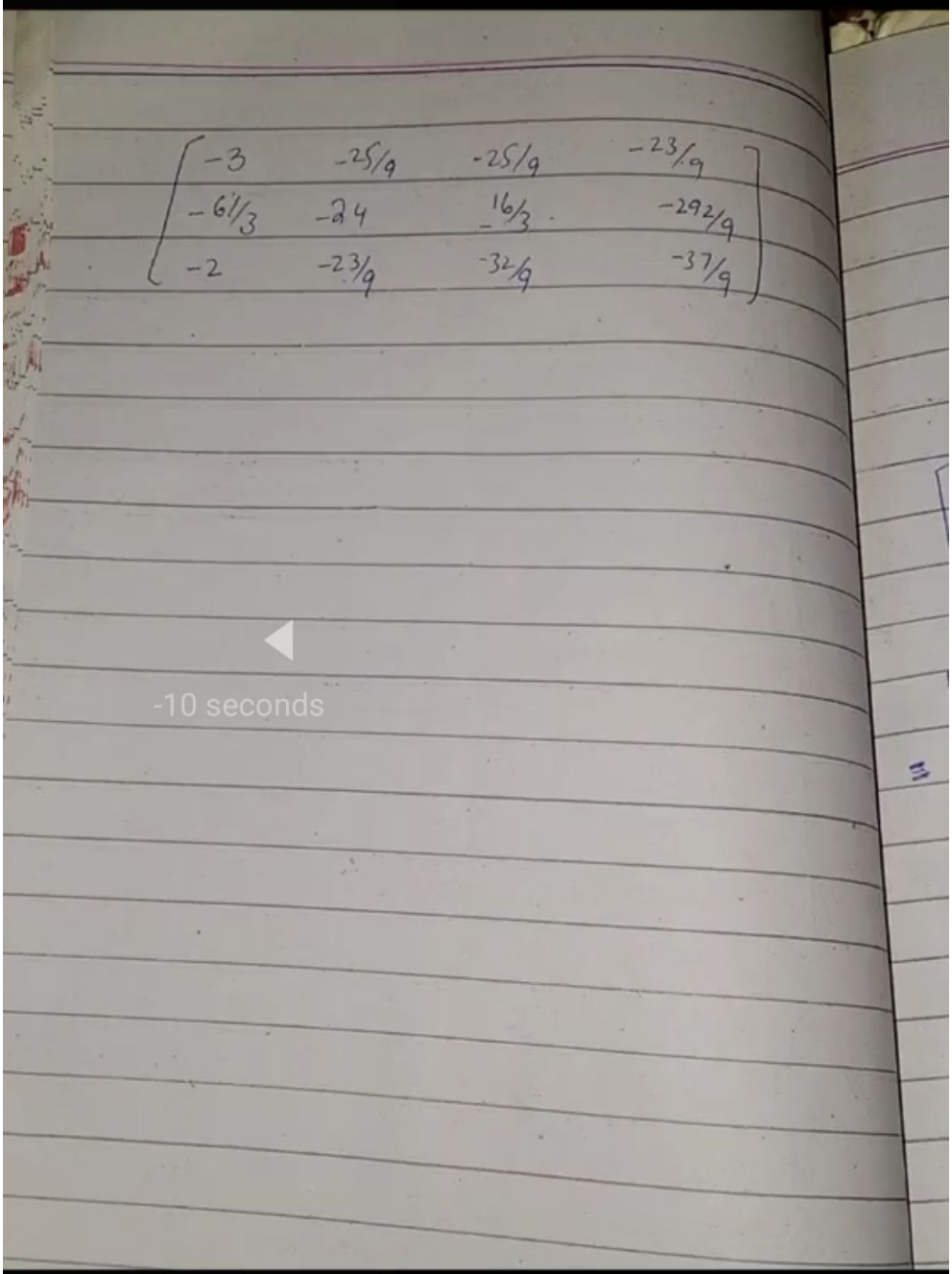
$$\begin{matrix} R_1 - R_2 \\ R_3 - 2R_2 \end{matrix} \begin{bmatrix} 1 & 0 & -5 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & -9 & 1 & 2 & 2 \end{bmatrix}$$

$$\frac{-1}{9} R_3 \begin{bmatrix} 1 & 0 & -5 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1/9 & -2/9 & -2/9 \end{bmatrix}$$

$$R_1 + 5R_3 \begin{bmatrix} 1 & 0 & 0 & -5/9 & -1/9 & -1/9 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1/9 & -2/9 & -2/9 \end{bmatrix}$$

$$R_2 - R_3 \begin{bmatrix} 1 & 0 & 0 & -5/9 & -1/9 & -1/9 \\ 0 & 1 & 0 & 1/9 & -7/2 & 2/9 \\ 0 & 0 & 1 & -1/9 & -2/9 & -2/9 \end{bmatrix}$$

$$A^{-1} \cdot B_2 \begin{bmatrix} -5/9 & -1/9 & -1/9 \\ 1/9 & -7/2 & 2/9 \\ -1/9 & -2/9 & -2/9 \end{bmatrix} \begin{bmatrix} 4 & 3 & 2 & 1 \\ 6 & 7 & 8 & 9 \\ 1 & 3 & 7 & 9 \end{bmatrix}$$


$\begin{bmatrix} -3 & -25/9 & -25/9 & -23/9 \\ -6\frac{1}{3} & -24 & -\frac{16}{3} & -292/9 \\ -2 & -23/9 & -32/9 & -37/9 \end{bmatrix}$
-10 seconds